

CSC212 Term 3: Practical 2

Due Date: 1 Oct 2026; 11:55PM

TOTAL: 50 marks

IMPORTANT: SUBMISSION IS VIA IKAMVA.

You must submit the Java Code, text file showing the output and a word/pdf document with time complexity analysis.

Question: A driver has been given the information in Table 1 to decide his schedule of trips for the month. As an expert in Algorithms and Complexity, your task is to help the driver decide on an optimal schedule based on the following constraints:

- He can only handle one car at a time.
- His commission is earned based on the maximum between number of trips he does in a day and the number of passengers he carries in a day.
- He can only drive for a 12-hour period in a day.
- He has to take a mandatory 30-minute break between trips.

To this end,

1. Implement, in Java, a Greedy approach for the maximum number of trips the driver can do in a day and a Dynamic Programming (Memorisation approach) for the maximum number of customers he can carry in a day.
2. Your code should output the answers to the following questions:
 - a. What is the optimal 12-hr window for the schedule from the Greedy approach?
 - b. What is his total commission if he earns R100 per trip at the end of the day based on the Greedy approach?
 - c. If he were to earn R10 per customer at every trip, what would be his commission if he selected an optimal schedule based on driving the largest number of customers in a day?
 - d. What are the cars that will be driven to earn this commission based on driving the largest number of customers in a day?
3. Submit a word/PDF document detailing the computational efficiency of your java code from question 1. It should analyse your loops, sorting algorithms, and dynamic array lookups and provide the Big-O Runtime for both the Greedy implementation and the Dynamic Programming block.

Table 1: Table showing each the schedule of trips for a day.

Car	Passenger Capacity	Trip Start time	Trip End time
A	10	08:00	16:00
B	50	01:00	08:00
C	42	13:00	21:00
D	20	22:00	23:59
E	13	02:00	04:00
F	20	07:00	12:00
G	10	15:00	19:00
H	35	17:00	23:59
I	15	05:00	10:00
J	45	11:00	14:00
K	30	00:01	06:00
L	18	20:00	23:00
M	20	13:00	18:00

Additional Instructions:

- An example of the output of your code (which should be saved in a .txt file) should be:

```
The optimal 12-hr window for the greedy approach is between X and Y
His total commission if he earns R100 per trip is RXXXX
His total commission if he were to earn R10 per customer is RXXXX
The cars to obtain the schedule based on largest customers is P, Q, R, S
```

- No input file is required; however, an *output.txt* file is required and must be submitted separately from the java implementation.
- Your algorithm implementation will be graded on the judicious use of comments, correctly incorporating the constraints as well as efficiency. Look carefully through the algorithms you have learnt so far in the course and pick the most efficient for this task.

Mark Allocation:

Question 1: 30 marks
Question 2: 10 marks
Question 3: 10 marks